EventTicketing — Core Domain / Entity Class Diagram
(khp úng EventTicketingSchema.sql — 26 bng — theo System design.docx)

Domain / Entity Layer

User
-id: UUID
-phone: String
-email: String
-fullName: String
-role: Enum{CUSTOMER,ORGANIZER,CEO,GATE_STAFF,ADMIN}
-status: Enum{PENDING,ACTIVE,LOCKED,DISABLED}
-phoneVerifiedAt: DateTime
+isEligibleToPurchase(): boolean

OtpVerification
-id: UUID
-purpose: String
-codeHash: String
-expiresAt: DateTime
-attemptCount: int
+verify(code: String): boolean

Venue
-id: UUID
-name: String
-address: String
-timezone: String

Event
-id: UUID
-name: String
-slug: String
-startsAt: DateTime
-endsAt: DateTime
-status: Enum{DRAFT,ON_SALE,SOLD_OUT,PAUSED,ENDED,CANCELLED}
-maxTicketsPerUser: int
+isOnSale(): boolean

VenueZone
-id: UUID
-code: String
-name: String
-capacity: int

Promotion
-id: UUID
-code: String
-discountType: Enum{PERCENT,FIXED}
-discountValue: Decimal
-usageLimit: int
-isActive: boolean

QueueEntry
-id: UUID
-sessionId: String
-position: long
-status: Enum{WAITING,ADMITTED,EXPIRED,LEFT,BLOCKED}
-botScore: Decimal

PriceChange
-id: UUID
-oldPrice: Decimal
-newPrice: Decimal
-reason: String
-source: Enum{MANUAL,RULE,AI,PROMOTION}
-changedAt: DateTime

BotChallenge
-id: UUID
-challengeType: String
-deviceFingerprint: String
-riskScore: Decimal
-result: Enum{PASSED,FAILED,BLOCKED}

AuditLog
-id: long
-action: String
-entityType: String
-entityId: UUID
-oldData: JSON
-newData: JSON

SalesSnapshot
-id: long
-capturedAt: DateTime
-ticketsSold: int
-ticketsAvailable: int
-grossRevenue: Decimal
-netRevenue: Decimal

ZonePriceRule
-id: UUID
-basePrice: Decimal
-floorPrice: Decimal
-ceilingPrice: Decimal
-isActive: boolean
+checkBounds(price: Decimal): boolean

GateDevice
-id: UUID
-deviceCode: String
-gateName: String
-lastSyncAt: DateTime
-isActive: boolean

Seat
-id: UUID
-rowLabel: String
-seatNumber: String
-seatLabel: String
-isActive: boolean

Order
-id: UUID
-orderCode: String
-subtotal: Decimal
-discountAmount: Decimal
-totalAmount: Decimal
-status: Enum{PENDING_PAYMENT,PAYMENT_PROCESSING,PAID,FAILED,CANCELLED,REFUND_PENDING,REFUNDED,PARTIALLY_REFUNDED}
-expiresAt: DateTime
+isExpired(): boolean
+calculateTotal(): Decimal

EventSeat
-id: UUID
-status: Enum{AVAILABLE,HELD,SOLD,BLOCKED}
-currentPrice: Decimal
-version: long
+lockAtomic(userId: String, duration: int): boolean

PromotionUsage
-id: UUID
-discountAmount: Decimal
-usedAt: DateTime

PaymentTransaction
-id: UUID
-provider: Enum{MOMO,VNPAY,SHOPEEPAY,CARD}
-providerTransactionId: String
-amount: Decimal
-currency: String
-status: Enum{INITIATED,PENDING,SUCCEEDED,FAILED,CANCELLED,REFUND_PENDING,REFUNDED}
+markAsPaid(): void

SeatHold
-id: UUID
-status: Enum{ACTIVE,EXPIRED,CONVERTED,CANCELLED}
-heldAt: DateTime
-expiresAt: DateTime
+release(): void

OrderItem
-id: UUID
-unitPrice: Decimal
-quantity: int

Refund
-id: UUID
-amount: Decimal
-reason: String
-status: String
+canRefund(): boolean

PaymentReconciliation
-id: UUID
-detectedAt: DateTime
-status: Enum{PENDING,MATCHED,TICKET_ISSUED,REFUND_TRIGGERED,RESOLVED,MANUAL_REVIEW}
-resolution: String

Ticket
-id: UUID
-ticketCode: String
-status: Enum{ISSUED,ACTIVE,USED,VOID,REFUNDED}
-issuedAt: DateTime
-usedAt: DateTime
+markAsUsed(): void
+isValid(): boolean

TicketQrToken
-id: UUID
-tokenHash: String
-validFrom: DateTime
-validUntil: DateTime
+isExpired(): boolean

TicketDelivery
-id: UUID
-channel: Enum{EMAIL,APP,APPLE_WALLET,GOOGLE_WALLET}
-destination: String
-status: String

Checkin
-id: UUID
-scannedAt: DateTime
-result: Enum{VALID,ALREADY_USED,INVALID,VOID,OFFLINE_CONFLICT}
-isOffline: boolean
-clientEventId: String
+resolveOfflineConflict(): void

approvedBy
changedBy
t chc (organizer)
din ra ti
staff
Cha khu vc
t mua
Phân b gh
Phiên bn gh theo s kin
Bao gm
Phát hành vé
Mã QR ng
Lch s soát vé

Ký hiu	Ý ngha
*--	Composition
o--	Aggregation
-->	Association mt chiu